ExtendingSiamese Open-Set Raman Experiments

Ordered Walkthrough, Results, and Interpretation

Repository Summary

March 2026

Study Goal

- Starting point: a pretrained Siamese Raman encoder trained on an older device and reference library.
- New challenge: a Feb. 26 acquisition set introduces both:
  - overlapping known classes across devices: benzene and pyridine
  - truly new classes: aniline, DCM, diethylamine, and n-hexane
- Main questions:
  - Can the frozen encoder classify the new data after alignment?
  - Can it reject truly unknown chemicals in an open-set setting?
  - Is failure driven by bad embeddings or by cross-device threshold mismatch?

Experiment Order

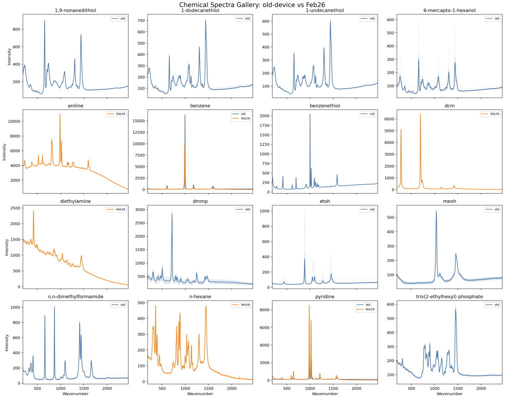
1. Align Feb. 26 spectra to the legacy Raman axis and test closed-set extension.
2. Run open-set detection with the frozen encoder and old-device reference only.
3. Recalibrate the rejection threshold using known spectra from both devices.
4. Fine-tune the encoder on overlapping known classes from both devices.
5. Re-run the open-set evaluation and compare the geometry before and after fine-tuning.

Data Preparation and Shared Pipeline

- Feb. 26 raw spectra were interpolated onto the old reference axis.
- Every spectrum then received:
  - ALS baseline correction
  - vector normalization
  - embedding through the same 1D-convolutional Siamese encoder
- Closed-set and open-set decisions were both made by nearest-neighbor search in embedding space.
- Distances are Euclidean distances between normalized embeddings.

1. Closed-Set Extension Setup

- One aligned Feb. 26 exemplar per class was appended to the reference.
- The rest of the Feb. 26 spectra were appended to the query set.
- The encoder stayed frozen.
- This asks a simple question: if the new classes are represented in the library, does the embedding separate them cleanly?



1. Closed-Set Extension Results

Query subset	Count	Top-1 acc.	Top-2 acc.
Expanded full query set	765	1.000	1.000
Original-label subset	669	1.000	1.000
Feb. 26 subset	200	1.000	1.000
True new labels only	96	1.000	1.000

- Interpretation: once one aligned exemplar per new class is added, the frozen embedding is already strong enough for perfect nearest-neighbor classification.
- This shows the embedding is chemically informative; it does not yet test open-set rejection.

2. Frozen-Encoder Open-Set Setup

- Reference remained the original old-device library only.
- Known query set: original old-device queries, $n = 565$.
- Unknown query set: Feb. 26 aniline, DCM, diethylamine, n-hexane, $n = 100$.
- Known controls: Feb. 26 benzene and pyridine, $n = 106$.
- For every query:
  - assign nearest reference label
  - compute nearest embedding distance
  - accept if distance $\leq t$, reject if distance $> t$

How the Open-Set Threshold Was Determined

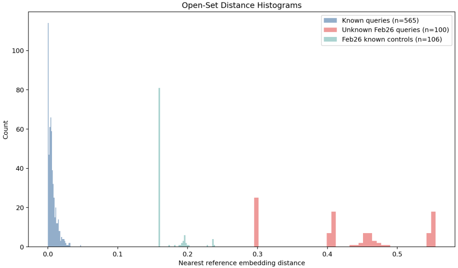
- Threshold selection was done by sweeping candidate values over the empirical distance distribution.
- Candidates were built from 401 quantiles of all nearest distances from:
  - old-device known queries
  - true unknown Feb. 26 queries
- For each threshold t, the script computed:
  - known-correct-and-accept rate
  - unknown reject rate
  - open-set accuracy
  - balanced accuracy $= 0.5 \times$ (known correct & accept $+$ unknown reject)
- The selected operating point was the threshold with:
  - highest balanced accuracy
  - then highest open-set accuracy
  - then smallest threshold among ties

2. Frozen-Encoder Open-Set Results

Metric	Value
Best threshold	0.14665
Known correct & accept	1.000
Unknown reject rate	1.000
Open-set accuracy	1.000
Balanced accuracy	1.000

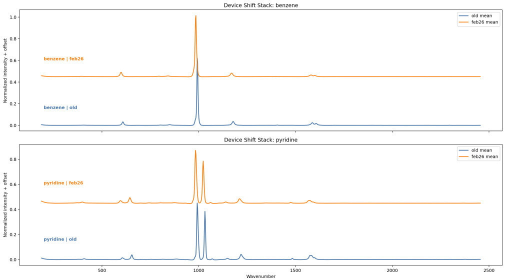
Distance summary:

- known mean: 0.00635, known max: 0.04765
- unknown mean: 0.42804, unknown min: 0.29514
- known-control mean: 0.16965



2. Interpretation of the Frozen-Encoder Result

- The threshold looks perfect only on the old-known vs true-unknown split.
- There is a wide empty gap between:
  - old known max distance: 0.04765
  - unknown min distance: 0.29514
- The chosen threshold 0.14665 is simply one candidate inside that gap.
- Feb. 26 benzene and pyridine expose the real failure:
  - top-1 class remained correct for both
  - but both were rejected at 100%
- So the issue was not class confusion. It was device-shifted distance inflation.



3. Threshold Recalibration Without Retraining

- The same frozen encoder was kept.
- The threshold sweep was repeated, but the known pool now included both:
  - original old-device known queries
  - Feb. 26 benzene and pyridine controls
- New best threshold: 0.23911.

Quantity	Frozen threshold	Recalibrated threshold
Best threshold	0.14665	0.23911
Unknown reject rate	1.000	1.000
Benzene reject rate	1.000	0.000
Pyridine reject rate	1.000	0.040

- Interpretation: the baseline embedding was mostly usable; the threshold was under-calibrated for cross-device knowns.

4. Cross-Device Fine-Tuning Setup

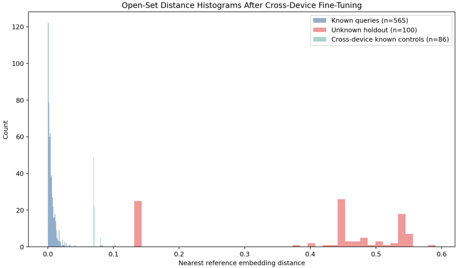
- Fine-tuning used the pretrained encoder as initialization.
- Added cross-device training data only for overlapping known classes:
  - 10 Feb. 26 benzene spectra
  - 10 Feb. 26 pyridine spectra
- The remaining Feb. 26 benzene and pyridine spectra were held out as controls.
- All true unknown classes remained fully held out.
- Training details:
  - contrastive loss on random positive/negative pairs
  - light augmentation: additive noise and small shifts
  - 40 epochs, learning rate 10^{-4}

4. Fine-Tuning Results

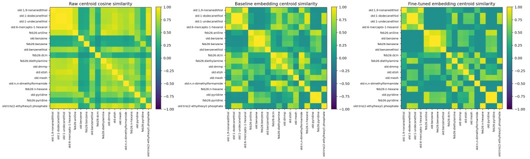
Metric	Value
Best threshold	0.12122
Known correct & accept	1.000
Known-control accept	1.000
Unknown reject rate	1.000
Open-set accuracy	1.000

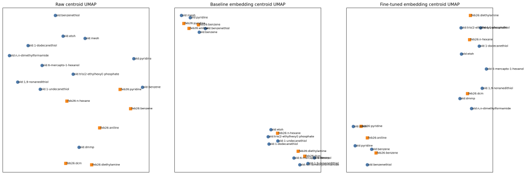
Control distances moved inward:

- benzene mean: $0.15930 \to 0.07003$
- pyridine mean: $0.20317 \to 0.08792$



Representation-Level Evidence From the Figures





Same-chemical cross-device centroid similarity:

- raw space: benzene 0.390, pyridine 0.454
- baseline embedding: benzene 0.988, pyridine 0.979
- fine-tuned embedding: benzene 0.998, pyridine 0.996

Overall Interpretation

- The pretrained encoder already captured the right chemical structure.
- That is why:
  - closed-set extension worked immediately
  - benzene and pyridine controls still got the correct nearest label
- The main deployment problem was absolute distance calibration across devices.
- Recalibration alone recovered most cross-device knowns.
- Fine-tuning on overlapping knowns solved the calibration problem more cleanly by pulling same-chemical cross-device embeddings closer together while keeping true unknowns separable.

Key Takeaways

1. Closed-set library extension is easy here: one aligned exemplar per new class was enough for perfect classification.
2. Open-set thresholding cannot be trusted if it is calibrated only on one device.
3. Cross-device known controls are essential; without them, a threshold can look perfect while failing in practice.
4. The frozen encoder was already good; the bottleneck was mostly cross-device distance shift.
5. Cross-device fine-tuning on overlapping known classes produced the cleanest operational result.